Problem Set 6 (Group 1)

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Question 1

In this question, we analyze data from a study by Sances (2016), which evaluated the effect of a town switching from an elected to an appointed tax assessor on the frequency of property tax reassessments in New York state during the 1980s through early 2000s.

Question 1a

```
# Load data
data <- read.csv("/Users/emrenono/Documents/UNC PhD/Spring 2026/POLI 784 (Causal)/Assignments/Assignmen

# Print number of unique units
num_units <- length(unique(data$swis_code))
glue("Number of units: {num_units}")
```

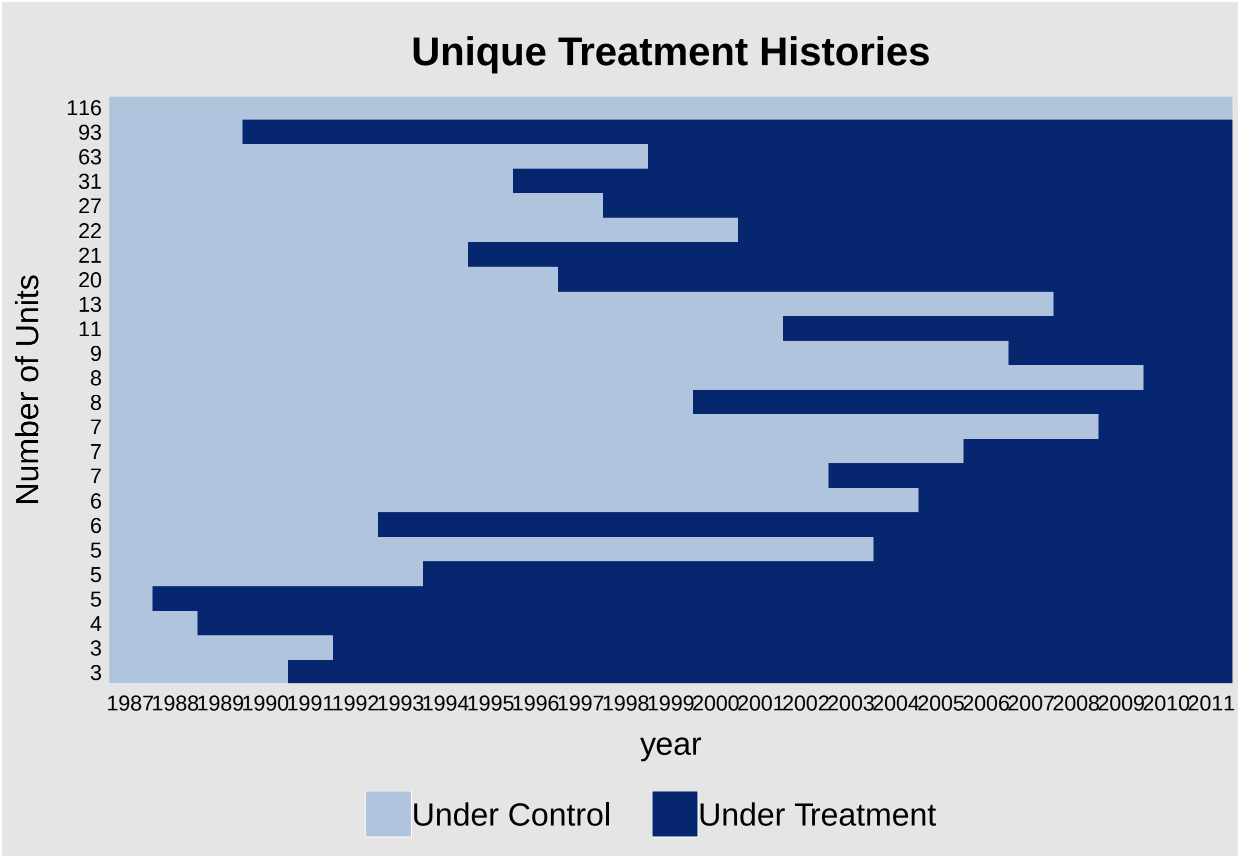
```
## Number of units: 510
```

```
# Print number of unique periods
num_periods <- length(unique(data$year))
glue("Number of periods: {num_periods}")
```

```
## Number of periods: 25
```

```
# Visualize treatment status using panelview
panelview(data=data,
          formula=reassessed~appointed,
          index=c("swis_code","year"))
```

```
## If the number of units is more than 300, we set "gridOff = TRUE".

## If the number of units is more than 500, we randomly select 500 units to present.
##         You can set "display.all = TRUE" to show all units.
```



Question 1b

Here, we estimate the effect of switching to an appointed assessor using a classic 2x2 DiD design that compares towns that switched in 1999 to those that never switched.

```r
# Filter data to include only towns that switched in 1999 (treated)
# and those that never switched (controls)
dat_DID <- data[data$cohort %in% c(1,14),]

# Get outcome variable
Y <- dat_DID$reassessed

# Create a treated/control indicator
treated <- (dat_DID$cohort == 14)
control <- !treated

# Create a pre/post treatment indicator
post_treatment <- (dat_DID$year >= 1999)
pre_treatment <- !post_treatment

# Calculate ATT using classic 2x2 DiD estimator
Ybar_treated_pre <- mean(Y[treated&pre_treatment])
Ybar_treated_post <- mean(Y[treated&post_treatment])
Ybar_control_pre <- mean(Y[control&pre_treatment])
```

```r
Ybar_control_post <- mean(Y[control&post_treatment])
ATT_1B <- (Ybar_treated_post - Ybar_treated_pre) - (Ybar_control_post - Ybar_control_pre)

glue("ATT estimate produced using classic 2x2 DiD estimator: {round(ATT_1B,6)}")
```

```
## ATT estimate produced using classic 2x2 DiD estimator: 10.680309
```

Question 1c

Here, we estimate the ATT using OLS regression of two-way transformed variables.

```r
# Create function to apply two-way transformation
two_way_transformation <- function(df,outcome_col,unit_col,period_col){

  # Get relevant columns
  df <- df[,c(outcome_col,unit_col,period_col)]
  colnames(df) <- c("Y","i","t")

  # Calculate unit- and period-specific means
  unit_means <- df %>% group_by(i) %>% summarise(unit_mean = mean(Y))
  period_means <- df %>% group_by(t) %>% summarise(period_mean = mean(Y))

  # Attach to dataframe
  df <- left_join(df,unit_means,by="i")
  df <- left_join(df,period_means,by="t")

  # Calculate grand mean
  grand_mean <- mean(df$Y)

  # Apply two-way transformation
  Y_tw <- df$Y - df$unit_mean - df$period_mean + grand_mean

  # Return result
  return(Y_tw)
}

# Apply transformation to outcome and treatment columns
dat_DID$reassessed_tw <- two_way_transformation(dat_DID,"reassessed","swis_code","year")
dat_DID$appointed_tw <- two_way_transformation(dat_DID,"appointed","swis_code","year")

# Estimate ATT via OLS
# (Because we applied two-way transformation, this should be equivalent to TWFE model)
mod_1C <- lm(reassessed_tw ~ appointed_tw, data = dat_DID)
ATT_1C <- mod_1C$coefficients["appointed_tw"]
summary(mod_1C)
```

```
##
## Call:
## lm(formula = reassessed_tw ~ appointed_tw, data = dat_DID)
##
## Residuals:
```

```
##     Min      1Q  Median      3Q     Max
## -66.923 -18.289  -6.923   9.102 111.205
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)  4.198e-17  4.645e-01   0.000        1
## appointed_tw 1.068e+01  1.947e+00   5.485 4.36e-08 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 31.33 on 4548 degrees of freedom
## Multiple R-squared:  0.006571,   Adjusted R-squared:  0.006353
## F-statistic: 30.08 on 1 and 4548 DF,  p-value: 4.36e-08
```

```
glue("ATT estimate produced using OLS regression and two-way transformation: {round(ATT_1B,6)}")
```

```
## ATT estimate produced using OLS regression and two-way transformation: 10.680309
```

The point estimates of the ATT produced in Part C are equivalent to those produced in Part B. This is reassuring since the regression in Part C is equivalent to fitting a two-way fixed effects (TWFE) model, which is equivalent to DiD in settings where all treated units receive treatment at the same time. Because we specifically filtered our data to include only towns that never switch (controls) and those that switch in 1999 (treated), this criteria is satisfied and we do not need to worry about "forbidden comparisons" or "negative weights."

Question 1d

Here, we use the feols package to fit a two-way fixed effects (TWFE) model. In the first model (shown below) we assume that errors are independently and identically distributed (i.i.d.) when computing standard errors, which is a common assumption in OLS regression.

```
## Model (1): i.i.d. errors (common assumption for OLS)
mod_1D_iid <- feols(
  fml = reassessed ~ appointed | swis_code + year,
  data = dat_DID,
  vcov = "iid"
)

# Extract ATT estimate and SE
ATT_1D_iid <- mod_1D_iid$coefficients["appointed"]
SE_1D_iid <- mod_1D_iid$se["appointed"]

# print results to console
summary(mod_1D_iid)
```

```
## OLS estimation, Dep. Var.: reassessed
## Observations: 4,550
## Fixed-effects: swis_code: 182,  year: 25
## Standard-errors: IID
##           Estimate Std. Error t value   Pr(>|t|)
## appointed  10.6803    1.99263  5.3599 8.7586e-08 ***
```

```
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
## RMSE: 31.3     Adj. R2: 0.274887
##              Within R2: 0.006571
```

```
glue("ATT estimate produced using feols with iid errors: {round(ATT_1D_iid,6)}")
```

```
## ATT estimate produced using feols with iid errors: 10.680309
```

```
glue("Standard error: {round(SE_1D_iid,6)}")
```

```
## Standard error: 1.992632
```

As expected, our point estimates of the ATT are equivalent to those produced in Parts B and C. However, because the error terms of individual units are likely to be autocorrelated (e.g., towns that conduct a reassessment in year t are unlikely to conduct another reassessment in year $t + 1$) the assumption of i.i.d. errors is likely violated. To overcome this issue, we can instead compute standard errors that are clustered at the unit-level.

```
## Model (2): clustered errors (more realistic for panel data)
mod_1D_clust <- feols(
  fml = reassessed ~ appointed | swis_code + year,
  data = dat_DID,
  vcov = cluster ~ swis_code
)

# Extract ATT estimate and SE
ATT_1D_clust <- mod_1D_clust$coefficients["appointed"]
SE_1D_clust <- mod_1D_clust$se["appointed"]

# print results to console
summary(mod_1D_clust)
```

```
## OLS estimation, Dep. Var.: reassessed
## Observations: 4,550
## Fixed-effects: swis_code: 182,  year: 25
## Standard-errors: Clustered (swis_code)
##           Estimate Std. Error t value  Pr(>|t|)
## appointed  10.6803    4.06961  2.6244 0.0094217 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
## RMSE: 31.3     Adj. R2: 0.274887
##              Within R2: 0.006571
```

```
glue("ATT estimate produced using feols with iid errors: {round(ATT_1D_clust,6)}")
```

```
## ATT estimate produced using feols with iid errors: 10.680309
```

```
glue("Standard error: {round(SE_1D_clust,6)}")
```

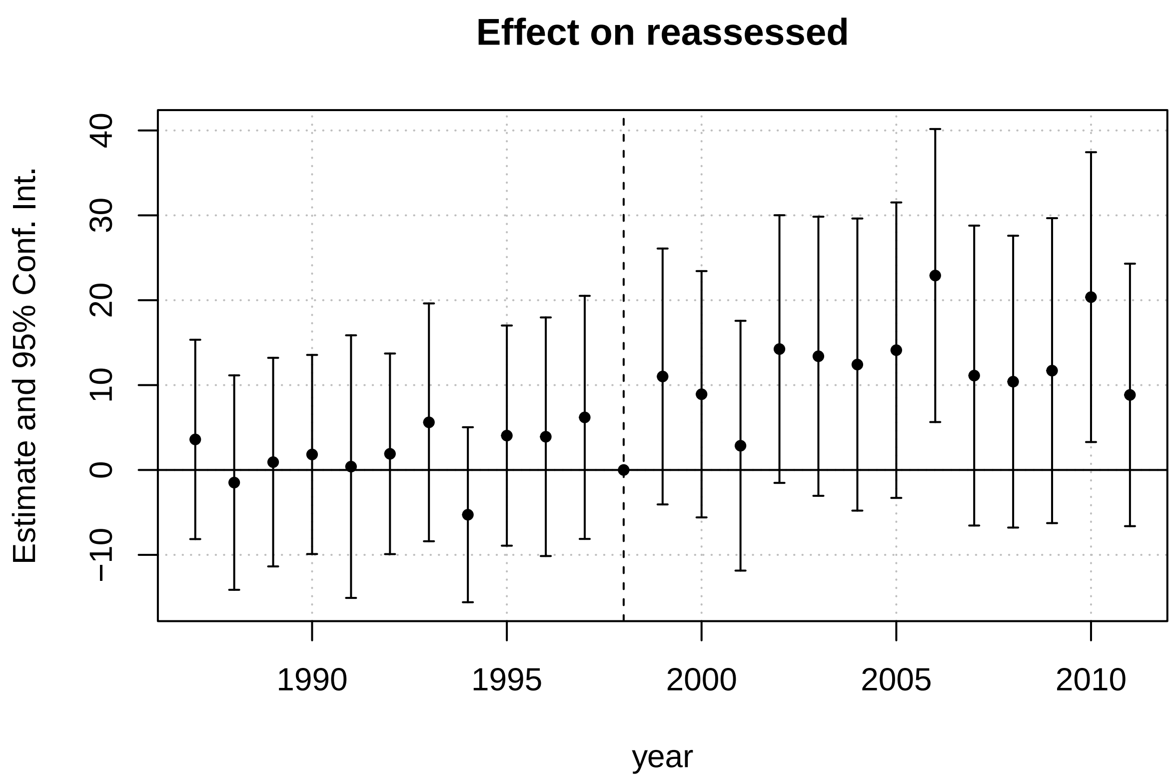
```
## Standard error: 4.069614
```

The above results suggest that standard OLS regression (which assumes i.i.d. errors) has a tendency to underestimate standard errors when autocorrelation is present. As such, the recommended practice is to cluster errors at the unit level when working with panel data.

Question 1e

Here, we fit a leads-and-lags model to check for violations of the parallel trends assumption.

```r
# Create binary variable indicating units that receive treatment
dat_DID$treated <- as.numeric(dat_DID$cohort==14)

# fit leads-and-lags model
mod_1E <- feols(
  fml = reassessed ~ i(year, treated, 1998) | swis_code + year,
  data = dat_DID,
  vcov = cluster ~ swis_code
)

# plot results
iplot(mod_1E)
```



As shown in the above plot, there does not appear to be a statistically-significant divergence in the coefficients of treated and control units during the pre-treatment period from 1987-1998. This suggests that the parallel trends assumption is not unreasonable in this scenario.

Question 1f

Here, we use feols to apply the TWFE estimator to our original dataset. This dataset contains a much larger sample of towns which can receive treatment at different points in time.

```r
# Fit TWFE model

mod_1F <- feols(
  fml = reassessed ~ appointed | swis_code + year,
  data = data,
  vcov = cluster ~ swis_code
)

# Extract ATT estimate and SE
ATT_1F <- mod_1F$coefficients["appointed"]
SE_1F <- mod_1F$se["appointed"]

# print results to console
summary(mod_1F)
```

```
## OLS estimation, Dep. Var.: reassessed
## Observations: 12,750
## Fixed-effects: swis_code: 510,  year: 25
## Standard-errors: Clustered (swis_code)
##           Estimate Std. Error t value   Pr(>|t|)
## appointed  8.61205    1.83572 4.69137 3.4918e-06 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
## RMSE: 33.8     Adj. R2: 0.266628
##              Within R2: 0.005053
```

```r
glue("ATT estimate produced TWFE model and original dataset: {round(ATT_1F,6)}")
```

```
## ATT estimate produced TWFE model and original dataset: 8.612045
```

```r
glue("Standard error: {round(SE_1F,6)}")
```

```
## Standard error: 1.83572
```

This approach is likely to result in biased estimates of the ATT because the TWFE estimator cannot accommodate staggered adoption of treatment; this occurs due to "forbidden comparisons" or "negative weights" that implicitly use already-treated units as controls for newly-treated units.

Question 1g

Here, we use feols to fit the following outcome model using only untreated observations:

$$Y_{it}(0) = \alpha_i + \gamma_t + \epsilon_{it} \forall (i,t) \notin \mathcal{M}$$

Where $\mathcal{M}$ is the set of treated observations. The fitted model is then used to impute the counterfactual outcome and individualistic treatment effect for treated observations:

$$\hat{\tau}_{it} = Y_{it} - \hat{Y}_{it}(0)$$

The ATT is subsequently estimated by taking an average of the individualistic treatment effects for treated observations:

$$A\hat{T}T = \frac{1}{|\mathcal{M}|} \sum_{(i,t) \in \mathcal{M}} \hat{\tau}_{it}$$

```
# Split data into set of treated / untreated observations
untreated_obs <- data[data$appointed == 0,]
treated_obs <- data[data$appointed == 1,]

# Fit TWFE model for untreated potential outcome
mod_1G <- feols(
  fml = reassessed ~ 1 | swis_code + year,
  data = untreated_obs,
  vcov = cluster ~ swis_code
)
```

```
## NOTE: 5/0 fixed-effect singletons were removed (5 observations).
```

```
# Use fitted model to impute counterfactual outcome for treated observations
treated_obs$reassessed_ct <- predict(mod_1G, newdata = treated_obs)

# Compute individualistic treatment effect (ITE)
treated_obs$ITE <- treated_obs$reassessed - treated_obs$reassessed_ct

# Compute ATT as average of ITEs
ATT_1G <- mean(treated_obs$ITE,na.rm=TRUE)

glue("ATT estimate produced via manual implementation of FEct estimator: {round(ATT_1G,6)}")
```

```
## ATT estimate produced via manual implementation of FEct estimator: 9.89246
```

Question 1h

Here, we use the fect package to estimate the following quantities: - The "average" ATT over the entire study period - The "dynamic" ATT stratified by the number of years since treatment adoption

```
# set seed for reproducibility
set.seed(42) # fect uses bootstrapping for variance estimation

# fit fixed-effects counterfactual estimator model
mod_1H <- fect(
  formula = reassessed ~ appointed,
  data = data,
  index = c("swis_code", "year"),
```

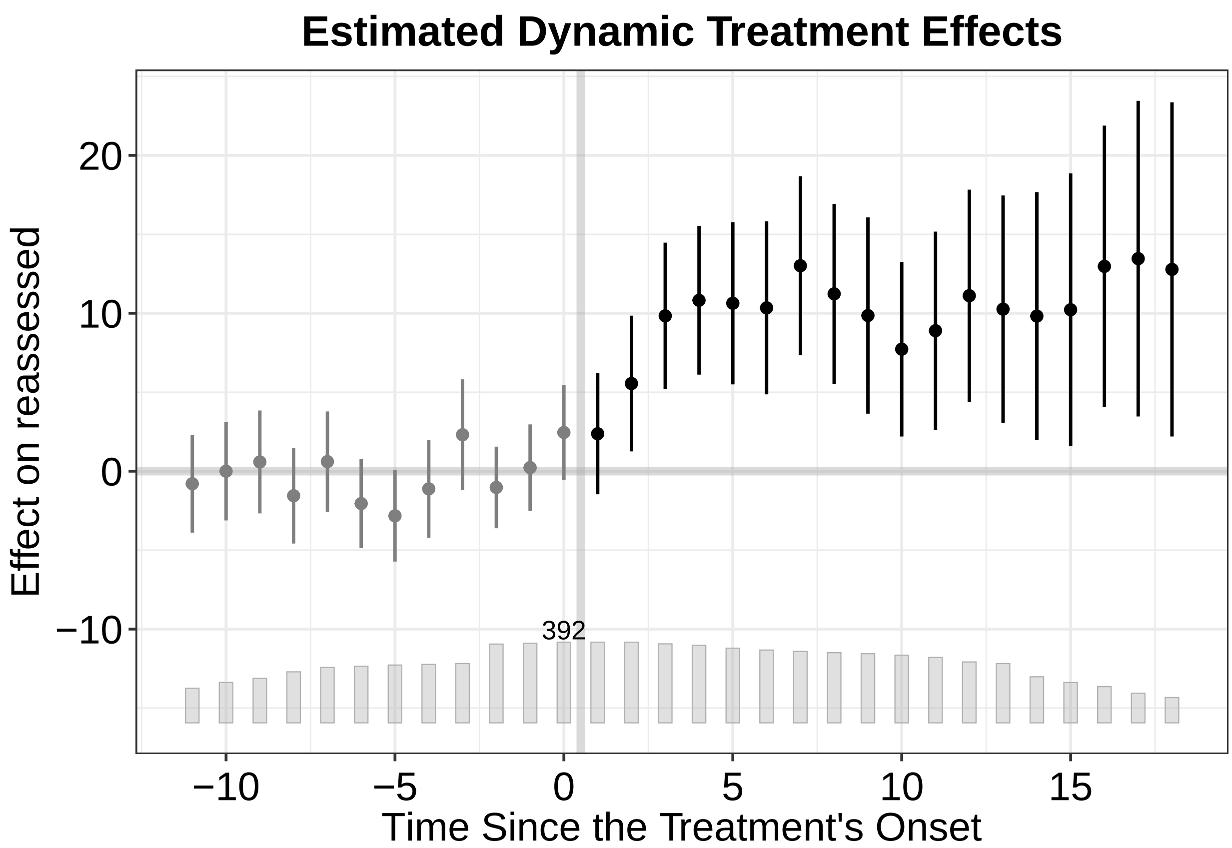
```r
  method = "fe",
  force = "two-way",
  se = TRUE,
  nboots = 1000
)
```

```
##
##  +--------------------------------------------------------------+
##  | Parallel computing: using 6 of 8 available cores.           |
##  |                                                              |
##  | To change: set cores = <n> in fect().                       |
##  | Default: min(available - 2, 8).                             |
##  +--------------------------------------------------------------+
```

```r
# print results to console
mod_1H
```

```
## Call:
## fect.formula(formula = reassessed ~ appointed, data = data, index = c("swis_code",
##     "year"), force = "two-way", method = "fe", se = TRUE, nboots = 1000)
##
## ATT:
##                               ATT  S.E. CI.lower CI.upper   p.value
## Tr obs equally weighted     9.722 2.071    5.663    13.78 2.673e-06
## Tr units equally weighted 10.761 2.106    6.634    14.89 3.211e-07
```

```r
# plot fect results
plot(mod_1H)
```



The above plot of dynamic treatment effects (which appear to be non-significant during pre-treatment periods) suggests that the estimator's assumption of no pre-trends might be reasonable. However, we can test this assumption in a more robust manner using placebo testing.

Question 1i

Here, we conduct a placebo test with three placebo periods to check for the presence of pre-trends.

```
# Conduct placebo test
mod_1I <- fect(
  formula = reassessed ~ appointed,
  data = data,
  index = c("swis_code", "year"),
  method = "fe",
  force = "two-way",
  se = TRUE,
  nboots = 1000,
  placeboTest = TRUE,
  placebo.period = c(-2, 0)
)
```

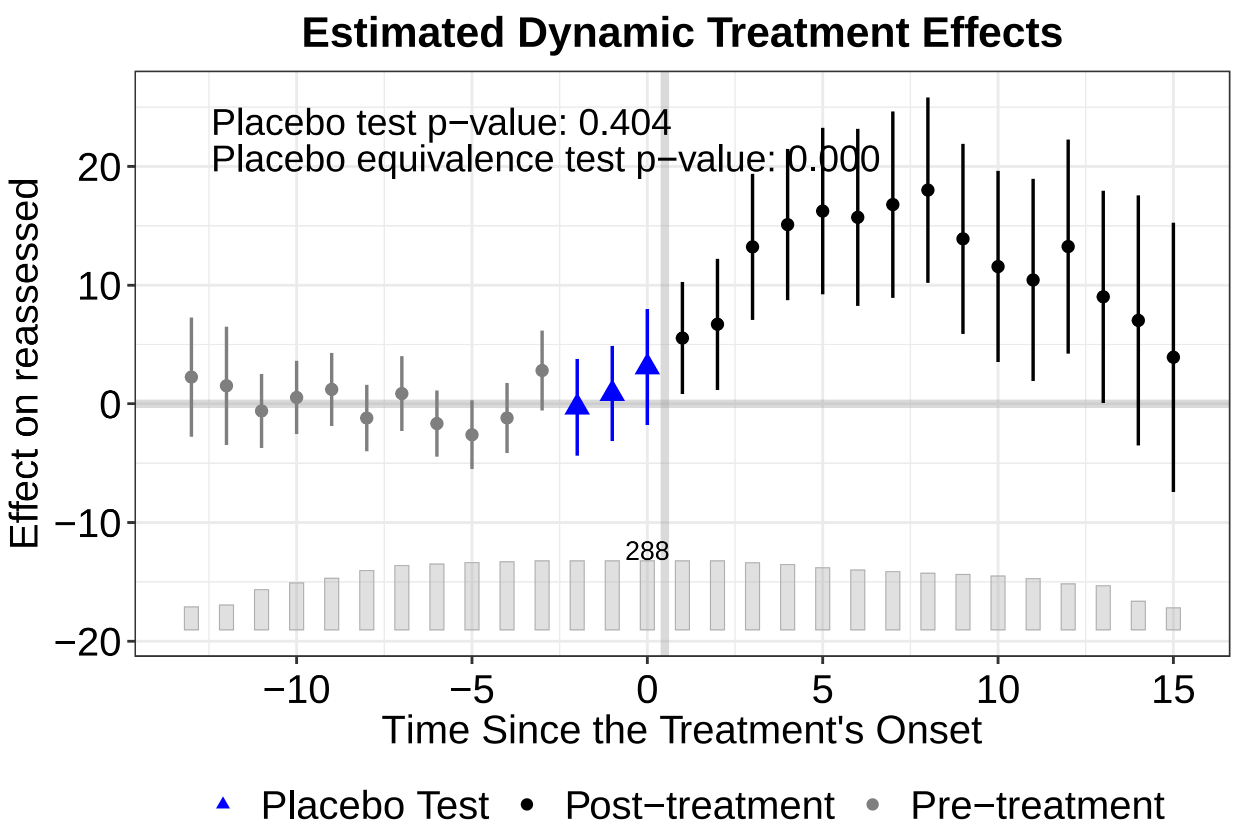
```
##
## +-----------------------------------------------------------+
## | Parallel computing: using 6 of 8 available cores.         |
```

```
##  |                                                     |
##  | To change: set cores = <n> in fect().               |
##  | Default: min(available - 2, 8).                     |
##  +-----------------------------------------------------+
```

```r
# print results to console
mod_1I
```

```
## Call:
## fect.formula(formula = reassessed ~ appointed, data = data, index = c("swis_code",
##     "year"), force = "two-way", method = "fe", se = TRUE, nboots = 1000,
##     placeboTest = TRUE, placebo.period = c(-2, 0))
##
## ATT:
##                             ATT  S.E. CI.lower CI.upper   p.value
## Tr obs equally weighted   12.13 2.743    6.757    17.51 9.726e-06
## Tr units equally weighted 12.77 2.769    7.342    18.20 4.005e-06
##
## Placebo effect for pre-treatment periods:
##      ATT.placebo S.E. CI.lower CI.upper p.value CI.lower(90%) CI.upper(90%)
## [1,]       1.228 1.47   -1.654     4.11  0.4037        -1.191         3.646
```

```r
# plot fect results
plot(mod_1I)
```



The built-in placebo test procedures of fect return two main diagnostic measures:

- The placebo test p-value. This p-value represents the probability of observing our effect size under the null hypothesis of $ATT = 0$ during pre-treatment placebo periods (i.e., no pre-trends). If this p-value is significant, then it indicates that the assumption of no pre-trends is violated.
- The placebo equivalence test p-value or "two one-sided tests" (TOST) p-value. This p-value represents the probability of observing near-zero effects during all pre-treatment placebo periods under the null hypothesis that the $ATT < -\theta_2$ or $ATT > \theta_1$ where $[-\theta_2, \theta_1]$ represents the range of values that the researcher deems to be sufficiently close to zero (i.e., the equivalence range). By default, fect sets this range as $[-0.36\hat{\sigma}_\epsilon, 0.36\hat{\sigma}_\epsilon]$, where $\hat{\sigma}_\epsilon$ is the estimated standard deviation of the residualized untreated outcome, but users can modify this parameter using the tost.threshold argument. If this p-value is significant, then it indicates that any pre-trends that might be present are small enough that they fall within the researcher's tolerance.

Our placebo test procedure found that p-value #1 was insignificant and p-value #2 was significant, both of which support the assumption of no pre-trends.

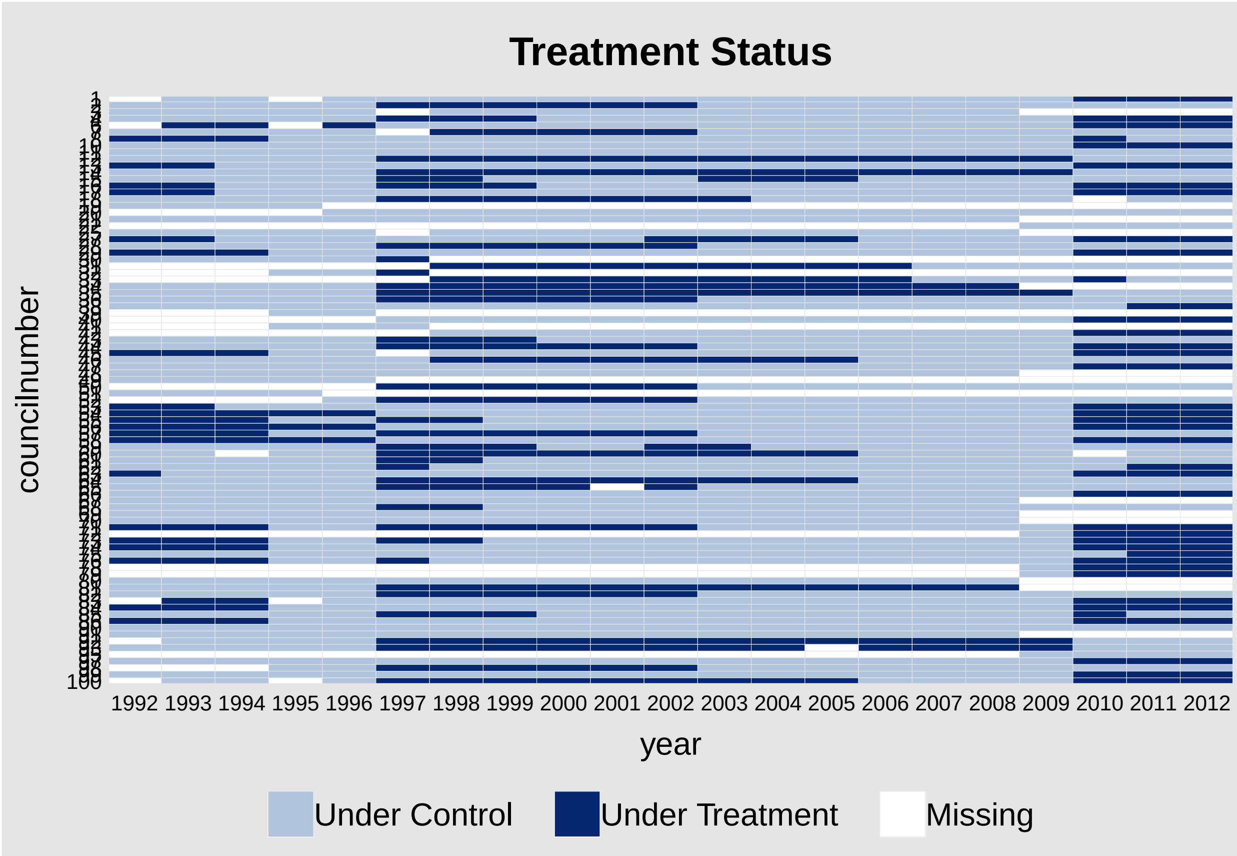
Question 2

This exercise is based on Fouirnaies and Mutlu-Eren (2015), who investigated whether partisan alignment between local councils and the central government in England bring localities more grants. The outcome of interest in the logarithm of specific grants per capita allocated to a local council (i.e., logSgwaPercap). The treatment is a dummy variable indicating whether the incumbent party controls the majority of the local council (i.e., treat). You can find the data needed to solve this exercise in data2.csv.

Question 2a

Use panelview() to visualize treatment status for the first 100 units in the dataset. Print the number of units (councilnumber) and periods (year) in the sample to console. Should we worry about the influence of carryover in this study?

```r
# Clear the environment
rm(list = ls())

# Load data into session
data2 <- read.csv("/Users/emrenono/Documents/UNC PhD/Spring 2026/POLI 784 (Causal)/Assignments/Assignme

# Use panelview() to visualize treatment status for the first 100 units in the dataset
panelview(
  data = data2 |>
    filter(councilnumber <= 100),
  formula = logSgwaPercap ~ treat,
  index = c("councilnumber", "year")
)
```




```r
# Print the number of units (councilnumber) and periods (year) in the same to console
num_units <- length(unique(data2$councilnumber))
num_periods <- length(unique(data2$year))
cat("Number of units (councilnumber):", num_units, "\n")
```

```
## Number of units (councilnumber): 432
```

```r
cat("Number of periods (year):", num_periods, "\n")
```

```
## Number of periods (year): 21
```

We should worry about the influence of carryover in this study because the treatment (partisan alignment) may have effects that persist over time, influencing future grant allocations even after the treatment status changes. This could lead to biased estimates of the treatment effect if not properly accounted for in the analysis.

Question 2b

Use the fixed effects counterfactual estimator from fect to estimate the ATT.

```r
# Set seed for reproducibility
set.seed(42)
```

```r
# Use the fixed effects counterfactual estimator from fect to estimate the ATT
att_estimate <- fect(
  formula = logSgwaPercap ~ treat,
  data = data2,
  index = c("councilnumber", "year"),
  method = "fe",
  se = TRUE,
  nboots = 1e3,
  seed = 42
)
```

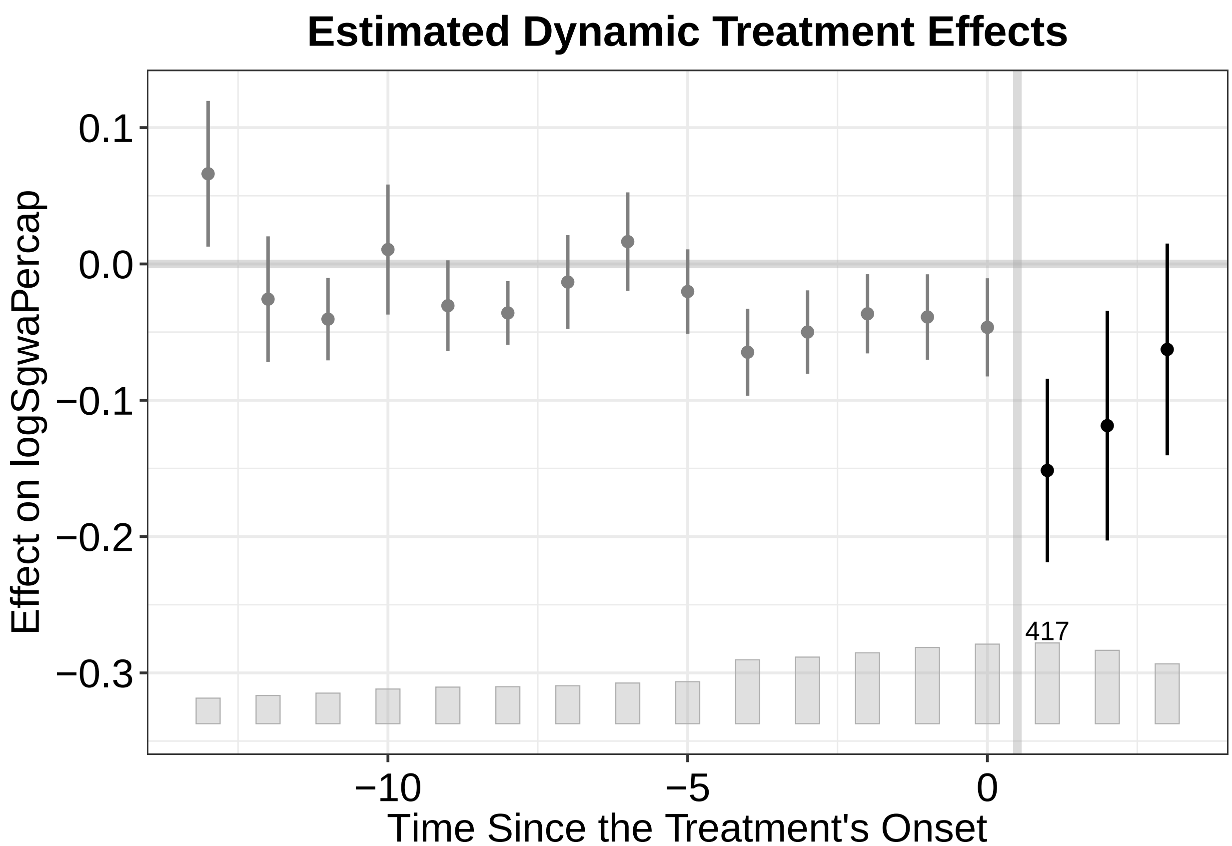
```
##
##  +-----------------------------------------------------------+
##  | Parallel computing: using 6 of 8 available cores.         |
##  |                                                           |
##  | To change: set cores = <n> in fect().                     |
##  | Default: min(available - 2, 8).                           |
##  +-----------------------------------------------------------+
```

i. Print the ATT point estimate to console and plot the ATT estimate for each period.

```r
# Print the ATT point estimate to console
print(att_estimate)
```

```
## Call:
## fect.formula(formula = logSgwaPercap ~ treat, data = data2, index = c("councilnumber",
##     "year"), method = "fe", se = TRUE, nboots = 1000, seed = 42)
##
## ATT:
##                                ATT    S.E. CI.lower  CI.upper p.value
## Tr obs equally weighted   -0.03831 0.02050 -0.07849  0.001871 0.06167
## Tr units equally weighted -0.05920 0.02951 -0.11704 -0.001363 0.04484
```

```r
# Plot the ATT estimate for each period
plot(att_estimate)
```

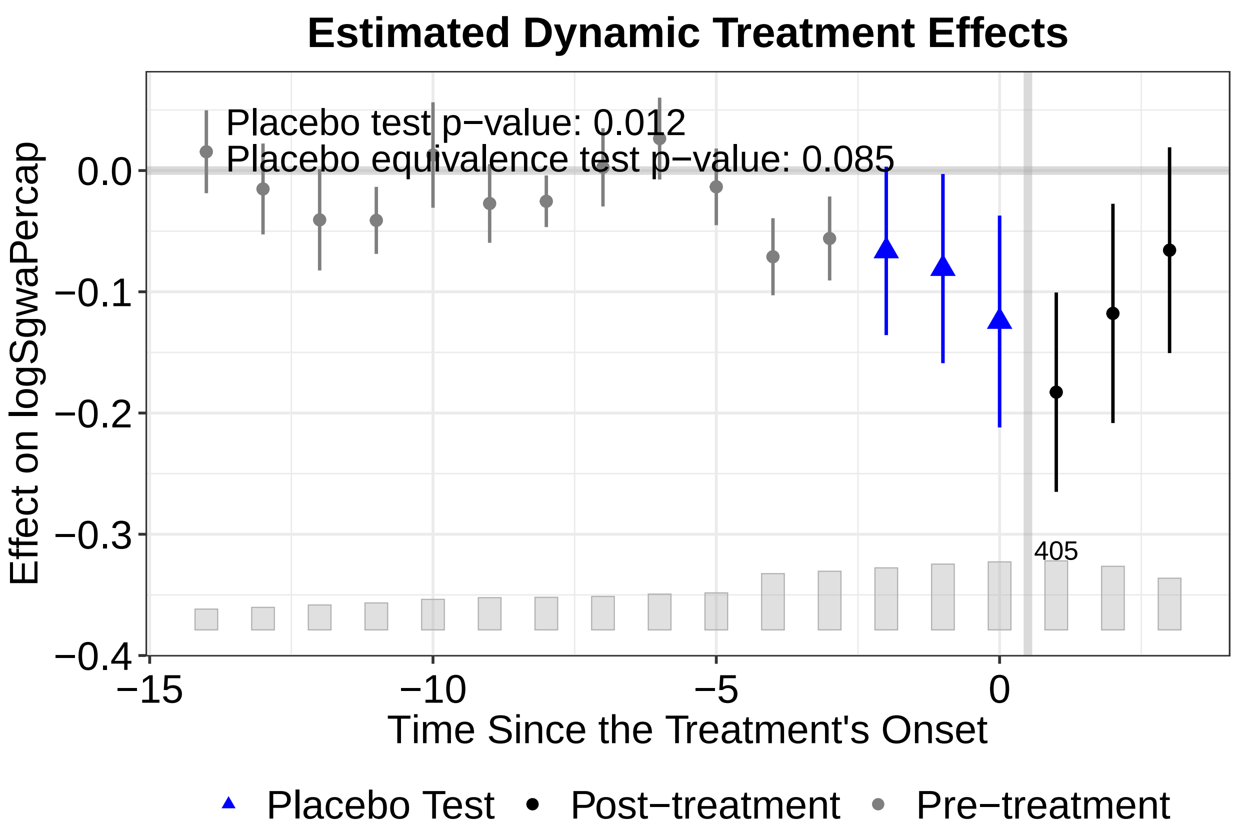


ii. Conduct a placebo test with the three placebo periods and plot the results.

```
# Set seed for reproducibility
set.seed(42)

# Conduct a placebo test with the three placebo periods
model_fe_placebo <- fect(
  formula = logSgwaPercap ~ treat,
  data = data2,
  index = c("councilnumber", "year"),
  method = "fe",
  se = TRUE,
  nboots = 1e3,
  seed = 42,
  placeboTest = TRUE,
  placebo.period = c(-2, 0) # specify the three placebo periods
)
```

```
##
## +-----------------------------------------------------------+
## | Parallel computing: using 6 of 8 available cores.         |
## |                                                           |
## | To change: set cores = <n> in fect().                     |
## | Default: min(available - 2, 8).                           |
## +-----------------------------------------------------------+
```

```r
# Plot the placebo results
plot(model_fe_placebo)
```



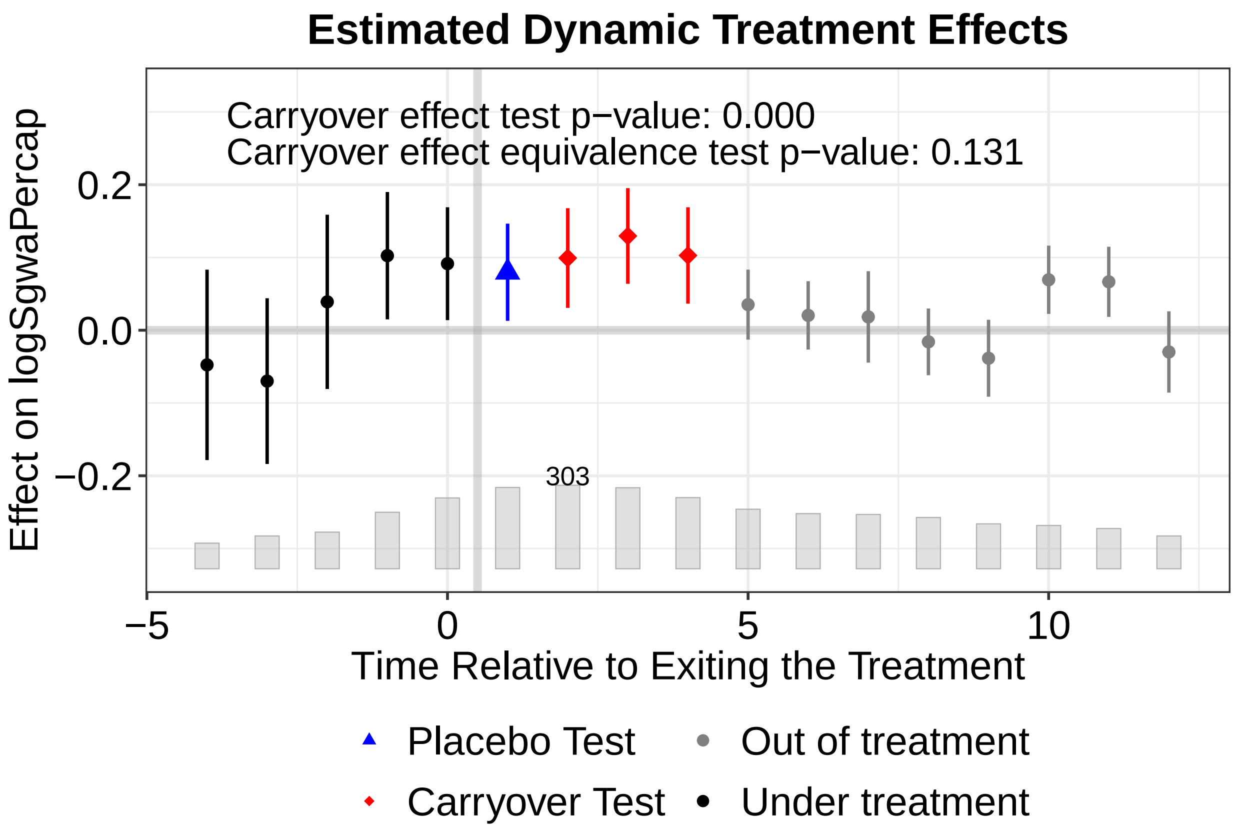
iii. Conduct a carryover test with three post-treatment periods and plot the results.

```r
# Set seed for reproducibility
set.seed(42)

# Conduct a carryover test with three post-treatment periods
model_fe_carryover <- fect(
  formula = logSgwaPercap ~ treat,
  data = data2,
  index = c("councilnumber", "year"),
  method = "fe",
  se = TRUE,
  nboots = 1e3,
  seed = 42,
  carryoverTest = TRUE,
  carryover.period = c(1, 3) # specify the three post-treatment periods
)
```

```
##
##  +-----------------------------------------------------------+
##  | Parallel computing: using 6 of 8 available cores.         |
##  |                                                           |
##  | To change: set cores = <n> in fect().                     |
```

```
##   | Default: min(available - 2, 8).                          |
##   +----------------------------------------------------------+
```

```r
# Plot the carryover test results
plot(model_fe_carryover)
```



iv. Do your results support the assumptions underlying the estimator?

The results do not support the assumptions underlying the fixed effects estimator. The placebo test is statistically significant (at p = 0.012), indicating the presence of pre-treatment effects and suggesting a violation of the no pre-trends assumption. Additionally, the placebo equivalence test is not significant, so the pre-treatment effects are not small enough to be considered negligible.

The carryover test is also statistically significant (at p = 0.001), showing evidence of pre-treatment effects. This suggests that the assumption of carryover is violated. Together, these results show that the assumptions of the fixed effects counterfactual estimator are not satisfied.

Question 2c

Use the interactive fixed effects counterfactual estimator from fect to estimate the ATT (NOTE: use the default number of factors).

```r
# Set seed for reproducibility
set.seed(42)

# Use the fixed effects counterfactual estimator from fect to estimate the ATT
att_estimate_ife <- fect(
  formula = logSgwaPercap ~ treat,
  data = data2,
  index = c("councilnumber", "year"),
  method = "ife",
  se = TRUE,
  nboots = 1e3,
  seed = 42
)
```

```
## For identification purposes, units whose number of untreated periods <5 are dropped automatically.
```

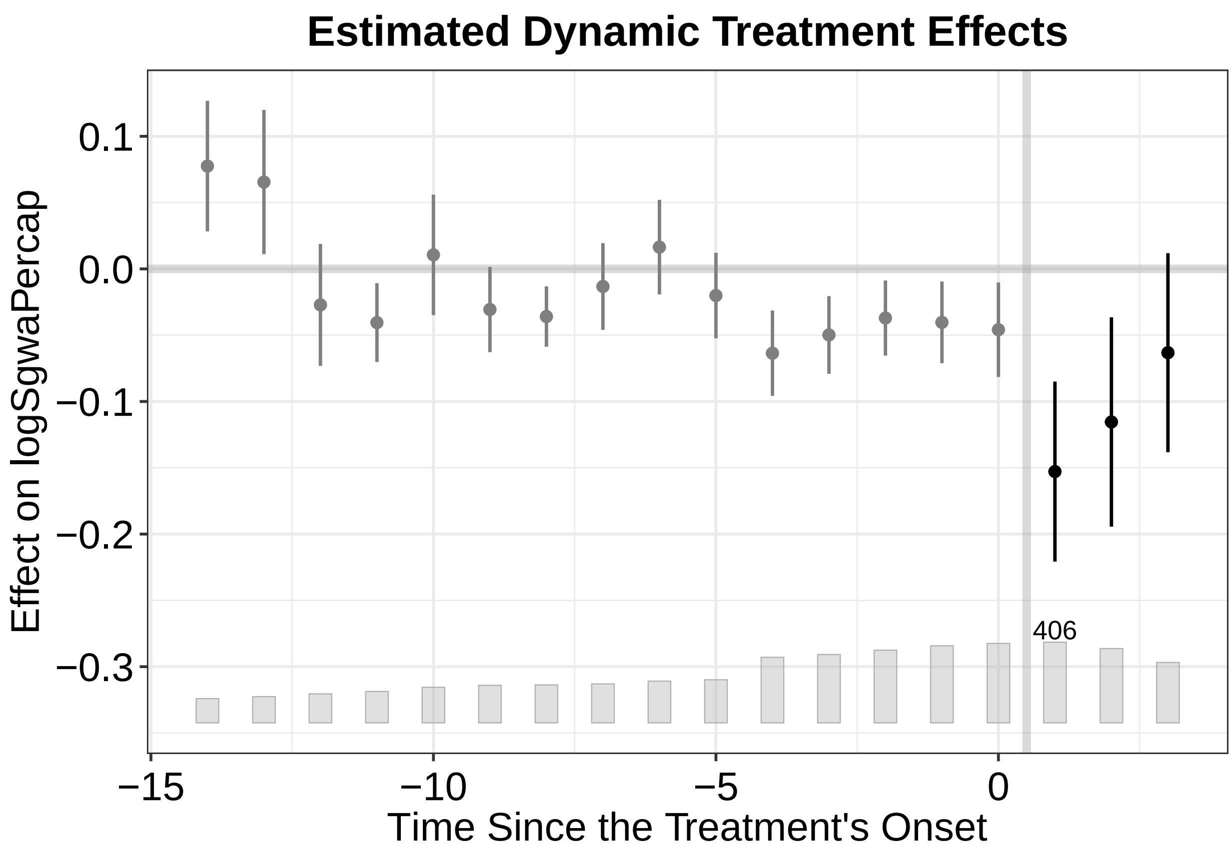
```
##
##  +--------------------------------------------------------------+
##  | Parallel computing: using 6 of 8 available cores.            |
##  |                                                              |
##  | To change: set cores = <n> in fect().                        |
##  | Default: min(available - 2, 8).                              |
##  +--------------------------------------------------------------+
```

i. Plot the ATT estimate for each period.

```r
# Print the ATT point estimate to console
print(att_estimate_ife)
```

```
## Call:
## fect.formula(formula = logSgwaPercap ~ treat, data = data2, index = c("councilnumber",
##     "year"), method = "ife", se = TRUE, nboots = 1000, seed = 42)
##
## ATT:
##                               ATT    S.E. CI.lower  CI.upper p.value
## Tr obs equally weighted   -0.03452 0.01986 -0.07344  0.004408 0.08221
## Tr units equally weighted -0.05867 0.02832 -0.11417 -0.003172 0.03827
```

```r
# Plot the ATT estimate for each period
plot(att_estimate_ife)
```

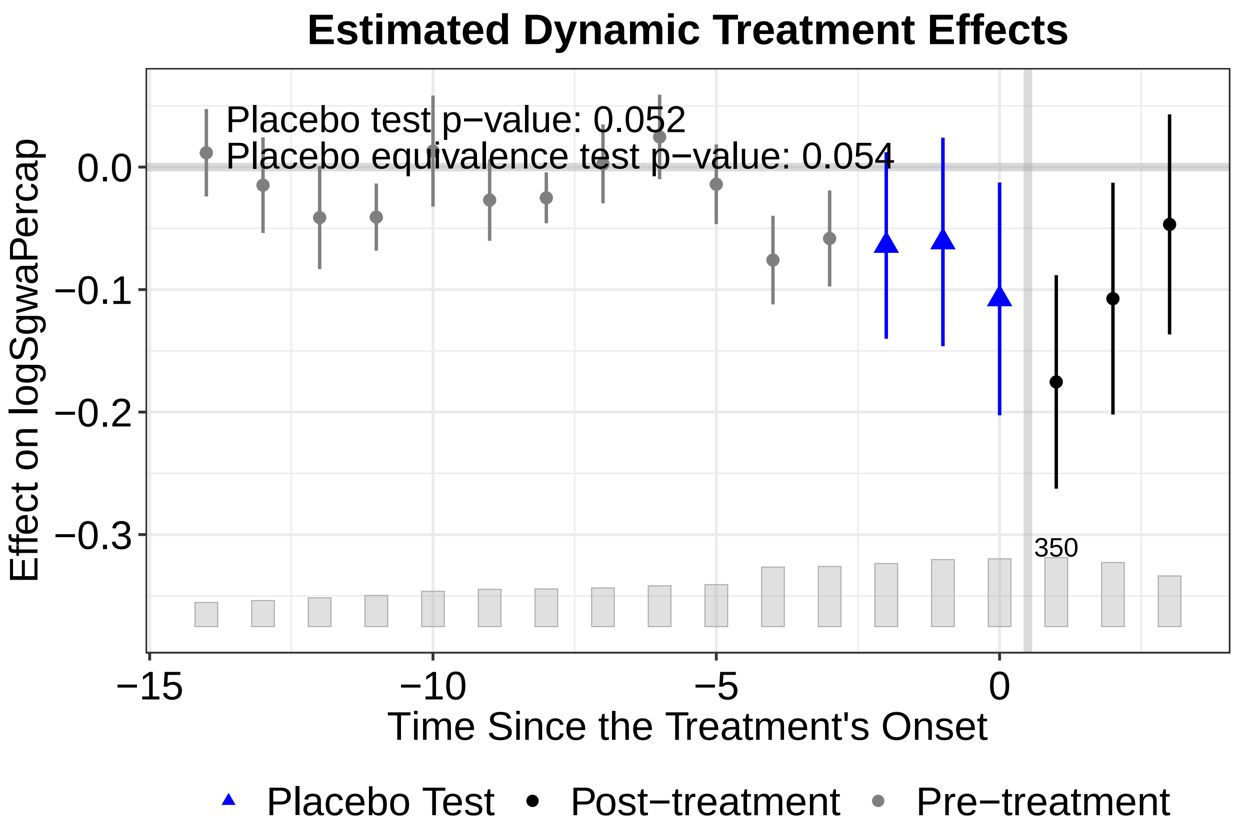


ii. Conduct a placebo test with three placebo periods and plot the results.

```
# Set seed for reproducibility
set.seed(42)

# Conduct a placebo test with the three placebo periods
model_ife_placebo <- fect(
  formula = logSgwaPercap ~ treat,
  data = data2,
  index = c("councilnumber", "year"),
  method = "ife",
  se = TRUE,
  nboots = 1e3,
  seed = 42,
  placeboTest = TRUE,
  placebo.period = c(-2, 0) # specify the three placebo periods
)
```

```
## For identification purposes, units whose number of untreated periods <5 are dropped automatically.

##
##   +-----------------------------------------------------------+
##   | Parallel computing: using 6 of 8 available cores.         |
##   |                                                           |
##   | To change: set cores = <n> in fect().                     |
##   | Default: min(available - 2, 8).                           |
##   +-----------------------------------------------------------+
```

```r
# Plot the placebo results
plot(model_ife_placebo)
```



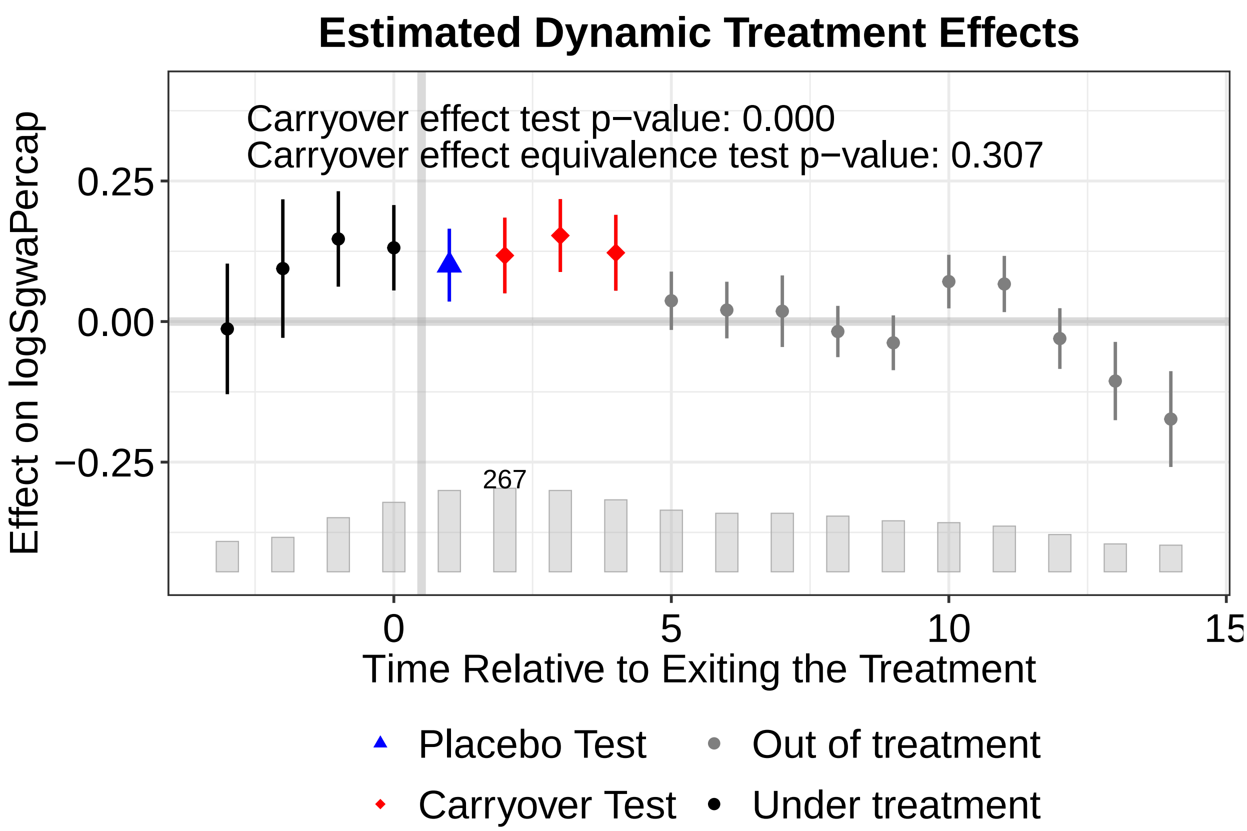
iii. Conduct a carryover test with three post-treatment periods and plot the results.

```r
# Set seed for reproducibility
set.seed(42)

# Conduct a carryover test with three post-treatment periods
model_ife_carryover <- fect(
  formula = logSgwaPercap ~ treat,
  data = data2,
  index = c("councilnumber", "year"),
  method = "ife",
  se = TRUE,
  nboots = 1e3,
  seed = 42,
  carryoverTest = TRUE,
  carryover.period = c(1, 3) # specify the three post-treatment periods
)
```

```
## For identification purposes, units whose number of untreated periods <5 are dropped automatically.

##
```

```
## +------------------------------------------------------------+
## | Parallel computing: using 6 of 8 available cores.          |
## |                                                            |
## | To change: set cores = <n> in fect().                      |
## | Default: min(available - 2, 8).                            |
## +------------------------------------------------------------+
```

```r
# Plot the carryover test results
plot(model_ife_carryover)
```



iv. Do your results support the assumptions underlying the estimator?

The results provide only partial support for the assumptions underlying the interactive fixed effects estimator. The placebo test yields a p-value of 0.052, which is slightly above the 0.05 significance threshold, and the placebo equivalence test is also borderline (p = 0.054). Together, these results suggest that there is limited but not definitive evidence of pre-treatment effects, making the validity of the no pre-trends assumption somewhat uncertain.

The carryover test is statistically significant (p = 0.000), indicating strong evidence of persistent post-treatment effects. This suggests that the assumption of no carryover is violated.

Overall, while the IFE estimator appears to better account for pre-treatment dynamics than the FE estimator, the presence of strong carryover effects raises concerns about the validity of its identifying assumptions. As a result, the ATT estimates should still be interpreted with caution.

Question 3

i. Regress educ on all other variables in the dataset using lm() to estimate the ATE of abd. Use age as the benchmark covariate for the sensitivity analysis.

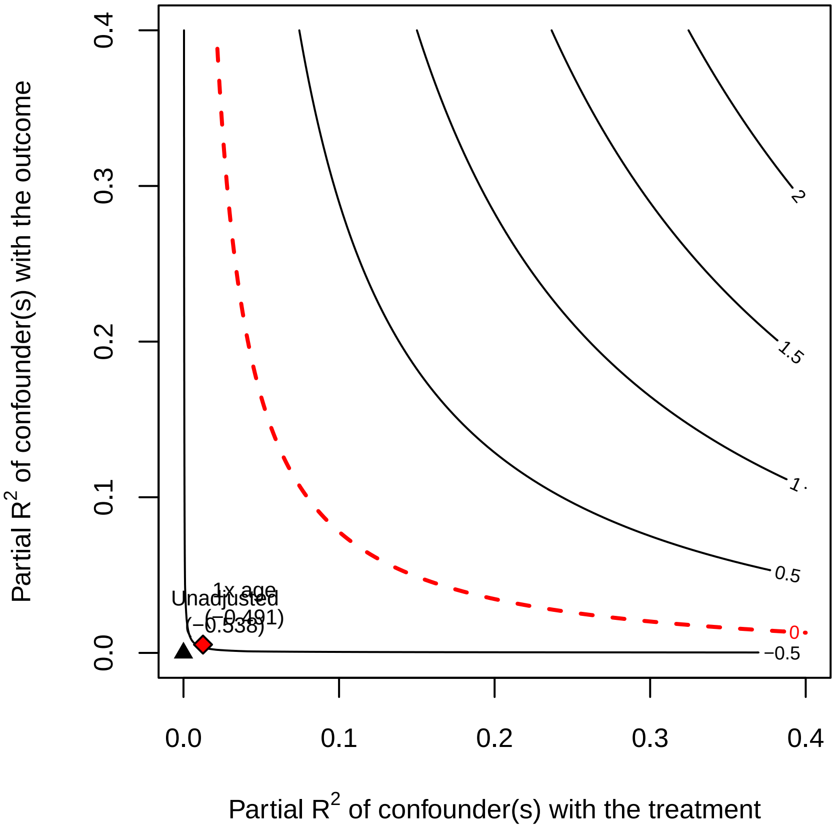
```r
# Load data into session
data3.2 <- read.csv("/Users/emrenono/Documents/UNC PhD/Spring 2026/POLI 784 (Causal)/Assignments/Assign

model3.1<- lm(data=data3.2, educ ~ abd + fthr_ed + mthr_ed + hh_size96 + orphan96 + hh_fthr_frm + age )
summary(model3.1)
```

```
##
## Call:
## lm(formula = educ ~ abd + fthr_ed + mthr_ed + hh_size96 + orphan96 +
##     hh_fthr_frm + age, data = data3.2)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -8.0301 -1.8662 -0.3445  1.9108  8.0252
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  5.24350    0.64964   8.071 2.85e-15 ***
## abd         -0.53824    0.21379  -2.518  0.01203 *
## fthr_ed      0.14820    0.03103   4.775 2.17e-06 ***
## mthr_ed      0.05475    0.03711   1.475  0.14055
## hh_size96    0.07364    0.02540   2.899  0.00385 **
## orphan96     0.29598    0.38681   0.765  0.44442
## hh_fthr_frm -0.37652    0.35941  -1.048  0.29516
## age          0.04059    0.02079   1.953  0.05123 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.788 on 733 degrees of freedom
## Multiple R-squared:  0.07717,    Adjusted R-squared:  0.06835
## F-statistic: 8.756 on 7 and 733 DF,  p-value: 2.321e-10
```

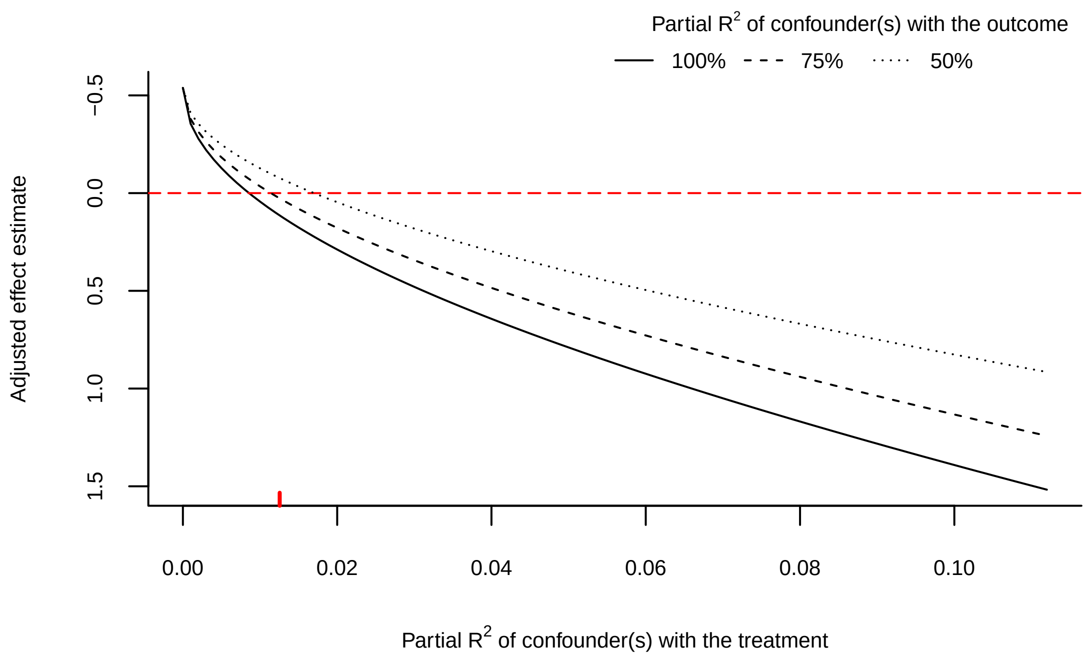
```r
# Use age as a benchmark for sensitivity analysis
sensitivity3.1 <- sensemakr(
  model = model3.1,
  treatment = "abd",
  benchmark_covariates = "age"
)
```

ii. Plot both the bias contour of the point estimate and the extreme scenario.

```r
plot(
  sensitivity3.1,
  type = "contour"
)
```




```
# Plot extreme scenario
plot(
  sensitivity3.1,
  type = "extreme"
)
```



iii. Print the summary of the sensitivity analysis to console and interpret your findings.

```r
print(sensitivity3.1)
```

```
## Sensitivity Analysis to Unobserved Confounding
##
## Model Formula: educ ~ abd + fthr_ed + mthr_ed + hh_size96 + orphan96 + hh_fthr_frm +
##     age
##
## Null hypothesis: q = 1 and reduce = TRUE
##
## Unadjusted Estimates of ' abd ':
##   Coef. estimate: -0.53824
##   Standard Error: 0.21379
##   t-value: -2.5176
##
## Sensitivity Statistics:
##   Partial R2 of treatment with outcome: 0.00857
##   Robustness Value, q = 1 : 0.08877
##   Robustness Value, q = 1 alpha = 0.05 : 0.02022
##
## For more information, check summary.
```

```r
summary(sensitivity3.1)
```

```
## Sensitivity Analysis to Unobserved Confounding
##
## Model Formula: educ ~ abd + fthr_ed + mthr_ed + hh_size96 + orphan96 + hh_fthr_frm +
##     age
##
## Null hypothesis: q = 1 and reduce = TRUE
## -- This means we are considering biases that reduce the absolute value of the current estimate.
## -- The null hypothesis deemed problematic is H0:tau = 0
##
## Unadjusted Estimates of 'abd':
##   Coef. estimate: -0.5382
##   Standard Error: 0.2138
##   t-value (H0:tau = 0): -2.5176
##
## Sensitivity Statistics:
##   Partial R2 of treatment with outcome: 0.0086
##   Robustness Value, q = 1: 0.0888
##   Robustness Value, q = 1, alpha = 0.05: 0.0202
##
## Verbal interpretation of sensitivity statistics:
##
## -- Partial R2 of the treatment with the outcome: an extreme confounder (orthogonal to the covariates)
##
## -- Robustness Value, q = 1: unobserved confounders (orthogonal to the covariates) that explain more t
##
## -- Robustness Value, q = 1, alpha = 0.05: unobserved confounders (orthogonal to the covariates) that
##
## Bounds on omitted variable bias:
##
## --The table below shows the maximum strength of unobserved confounders with association with the trea
##
##  Bound Label R2dz.x R2yz.dx Treatment Adjusted Estimate Adjusted Se Adjusted T
##       1x age 0.0125  0.0053       abd           -0.4906      0.2147    -2.2848
##  Adjusted Lower CI Adjusted Upper CI
##            -0.9121           -0.0691
```

These results show us that if the unobserved confounders explained the remaining variance of the outcome, they would have to explain AT LEAST 0.86% of residual variance to make the observed effect zero. If the unobserved confounders explained 8.88% of the treatment AND outcome, they would drive the effect to zero. If unobserved confounders explained 2.02% of the variance of BOTH the treatment and outcome, the estimate would be statistically not significant at the 95%-confidence level.

The summary statistic shows us that the benchmark covariate age would not reach any of the thresholds above to make the point estimate zero or statistically insignificant. So, if age is a typical covariate, then we are not as concerned about unobserved confounders biasing the results.